# Limitations and claim boundaries

The boundaries here define the contribution. The central goal is to show — concretely and testably — that the categorical generalized-variational construction inspired by FedGVI (Mildner et al. 2025) has standard-Bayes client limits and a project-local log-linear-pool server identity.

# Limitations and claim... (part 2)

Under the explicit bridge in sec. 8.8, the latter specializes Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024), not the complete source protocol; behavior away from those limits is evaluated only under declared simulation conditions.

# Limitations and claim... (part 3)

The exact identities are formal; performance and deployment claims remain conditional. Items beyond that boundary are named as future work.

# Three robustness axes: theorem, heuristic, and objective

The authoritative definitions and claim owners are in sec. 6.3; this section records only what still cannot transfer. A source-conditional client theorem does not certify either server rule,

# Three robustness axes (part 2)

a server recovery identity does not supply an objective or estimator-level influence bound, and objective descent plus raw-weight control does not imply truth recovery or peak-accuracy dominance. Likewise,

# Three robustness axes (part 3)

conditional accuracy contrasts cannot confer a theorem on the heuristic. The evidence map in fig. 9 makes that no-transfer boundary and its nested replication units explicit.

# Three robustness axes (part 4)

The remaining methodological problem is a server objective that combines competitive conditional performance with an appropriate robustness guarantee. Related robust-federation work motivates the target (Blanchard et al. 2017; Li et al. 2022) but does not certify either project server rule.

# Three robustness axes (part 5)

Any such result must also survive beyond the present categorical model, declared attack grid, and single-host execution setting; future work and extended methods remain separate (sec. 11.17, sec. 16).

# Scope boundaries that the evidence does not cross

**The bridge is a recasting, not an upstream claim.** Friston et al. (Friston et al. 2024) supply a belief-sharing operator for agents with a shared world model; Mildner et al.

# Scope boundaries that the... (part 2)

(Mildner et al. 2025) supply robust federated generalized variational inference. Neither paper claims the other.

# Scope boundaries that the... (part 3)

The contribution here is to recast the former as the tested KL/NLL zero-robustness recovery corner of the latter-inspired construction, then measure changes under robust losses and server reweighting.

# Scope boundaries that the... (part 4)

**Historical sources are conceptual, not formal support.** The pre-modern sources added in sec. 11.9 support a genealogy of expectation, inverse inference, utility, and collective judgment (Huygens 1657; Bayes 1763; Laplace 1774; Bernoulli 1738; Condorcet 1785). They do not support claims about KL, NLL,

# Scope boundaries that the... (part 5)

product-of-experts training, FedGVI, or `robust_aggregate`.

Those claims rest only on the modern cited formalism and the tests reported in sec. 10.1.

# Scope boundaries that the... (part 6)

**Single-machine federation.** Federation is validated with queue transport, a single-machine OS-process helper, and loopback TCP: agents serialize beliefs, the server aggregates, and consensus is broadcast back without changing the mathematics.

# Scope boundaries that the... (part 7)

The socket path exercises frame integrity and file-backed digest-verified replay validation; an optional SQLite round-ID guard survives local process restarts.

# Scope boundaries that the... (part 8)

These controls do not substitute for cross-host deployment, identity-bound mTLS, a shared multi-host replay domain, discovery, long-running worker orchestration, or fault tolerance; those steps are scoped as future work (sec. 11.17).

# Scope boundaries that the... (part 9)

**Source-scale posterior FedGVI classification remains open.** The displayed NumPy logistic-regression baseline is an exploratory point-estimate comparison, not the source paper's posterior-uncertainty BNN experiment (Mildner et al. 2025).

# Scope boundaries that the… (part 10)

Its operating point was selected within the evaluated synthetic sweep to expose a mid-contamination margin (fig. 21).

# Scope boundaries that the... (part 11)

The curves lose reliable ordering at the highest rate. The source theorem, not this selected finite sweep, carries the source-conditional robustness claim.

# Scope boundaries that the... (part 12)

**Classification baselines are point estimates, not full posteriors.** The NumPy logistic-regression baseline and the PyTorch MLP complement both use point-estimate weights; no posterior covariance is computed.

# Scope boundaries that the... (part 13)

A genuine mean-field variational family over the weights—diagonal-Gaussian $q(w)$ with a closed-form KL and a Monte-Carlo ELBO—is implemented by the tested `VariationalMLP` primitive in the `bnn_variational_torch` module. It recovers the point-estimate net exactly as its posterior variance vanishes.

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# Scope boundaries that the... (part 14)

The source-comparable training and evaluation lane remains unexecuted: leakage-free calibration, source-dataset parity, posterior-family comparisons, and source-scale compute are still open.

# Scope boundaries that the... (part 15)

**Hierarchical depth does not by itself improve the base task.** The two- and three-level stacks (sec. 17.9, sec. 17.11) run alternating L1/L2(/L3) inference end-to-end and resolve their added context latents above chance. On the shared location task they do not beat the flat baseline:

# Scope boundaries that the… (part 16)

the paired location-accuracy gap is a small, statistically reliable negative gap at the reported seed count (fig. 31).

# Scope boundaries that the... (part 17)

Depth is therefore validated as executable and consensus-preserving, not as an accuracy improvement; whether a richer policy-and-horizon task family rewards hierarchy is future work (sec. 11.20).

# Scope boundaries that the... (part 18)

**Discrete POMDP only—continuous or hybrid state spaces are not addressed.** This work is discrete-categorical only, matching the community's worked POMDP example (Da Costa et al. 2020; Friston et al. 2024).

# Scope boundaries that the… (part 19)

The limit-as-proof contract is validated off the categorical case only for a one-dimensional Gaussian-mean conjugate slice; continuous-state active inference and Gaussian belief-sharing colonies remain untested.

# Scope boundaries that the... (part 20)

**Temperature and divergence calibration are fixed, not learned.** Coarsened posterior and safe-Bayes theory make clear that the learning-rate/temperature is part of the inference rule under misspecification (Miller and Dunson 2018; Grünwald 2012; Kleijn and Vaart 2012).

# Scope boundaries that the… (part 21)

This manuscript validates the configured losses and divergences, but it does not learn an optimal coarsening radius, temperature, or divergence schedule across agents.

# Scope boundaries that the... (part 22)

**Real multi-machine federation, networking, and privacy cryptography.** Federation here is *mathematical* (factor aggregation), not infrastructural— unlike communication-efficient or Byzantine-robust federated learning (McMahan et al. 2017; Blanchard et al. 2017).

# Scope boundaries that the… (part 23)

**New linguistic theory.** The language-acquisition study (sec. 10.3) reproduces the Dirichlet count mechanism (eq. 18) mechanically; it does not extend the linguistics.

# What the statistics can and cannot claim

The verdict is a paired comparison at a single high contamination rate (sec. 10.5.2, tbl. 7). The design concentrates power where the effect is largest.

# What the statistics can and... (part 2)

It therefore identifies *where* robustness pays off, but it does not characterize contamination as a continuous function.

# What the statistics can and... (part 3)

The per-rate effect and inference projections (tbls. 5, 6) describe the remaining sweep.

# What the statistics can and... (part 4)

Their inferential claims do not exceed the declared BH-FDR control (Benjamini and Hochberg 1995).

# What the statistics can and... (part 5)

BH-FDR controls expected false discovery proportion within the declared family, not the chance of any false positive. The matched-pairs Wilcoxon tests are rank tests under paired-difference assumptions (Wilcoxon 1945; Fay and Proschan 2010), not assumption-free proofs about raw means.

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# What the statistics can and... (part 6)

Confidence intervals are percentile bootstrap (Efron and Tibshirani 1993), not analytic, and inherit that method's small-sample caveats at the lowest trial counts.

# What the statistics can and... (part 7)

The power values are observed-effect design approximations useful for confirmatory sample-size planning; they are not independent evidence for the verdict and do not cover model-specification uncertainty outside the seeded simulation harness (Wasserstein and Lazar 2016).